

CSC - 321

DESIGN AND ANALYSIS OF ALGORITHMS



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Matrix-Chain Multiplication

Matrix Chain Multiplication

Dot Product

$$\begin{bmatrix} 1 & 5 & 9 & 7 & 3 & 4 \\ 2 & 1 & 9 & 7 & 2 & 6 \\ 9 & 5 & 2 & 2 & 3 & 5 \\ 6 & 6 & 1 & 3 & 1 & 7 \end{bmatrix} \times \begin{bmatrix} 5 & 1 & 3 \\ 9 & 5 & 1 \\ 8 & 7 & 6 \\ 9 & 6 & 8 \\ 8 & 1 & 3 \\ 2 & 2 & 9 \end{bmatrix} = \begin{bmatrix} 217 & 142 & 163 \\ 182 & 126 & 177 \\ 158 & 73 & 114 \\ 141 & 76 & 120 \end{bmatrix}$$

4x6 6x3 4x3

72 Multiplications
in Total!
(4x6x3)

Matrix Chain Multiplication

$$\begin{array}{ccccccccc} A_1 & \times & A_2 & \times & A_3 & \times & A_4 & \times & A_5 \\ 4 \times 10 & & 10 \times 3 & & 3 \times 12 & & 12 \times 20 & & 20 \times 7 \end{array}$$

Goal: Find the optimal way to multiply these matrices to perform the fewest multiplications.

Naïve Approach: Try them all, and pick the most optimal one.

Running time: $\Omega(4^n/n^{3/2})$ - 4^n dominates! Exponential

Matrix Chain Multiplication

Now we want to try out a bunch of values for 'k' in order to see what the best one is:

$$M[i,j] = M[i,k] + M[k+1,j] + p_{i-1}p_kp_j$$

100 200 2x3x4

Since we don't know what k is, we try this range of k:

$$\begin{matrix} 100 & 200 \\ (A_{i\dots k}) & (A_{k+1\dots j}) \\ 2 \times 3 & 3 \times 4 \end{matrix}$$

The minimum returned value is our solution!

$$i \leq k < j$$

Our Final Recursive Formula:

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} & \text{otherwise} \end{cases}$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} & \text{otherwise} \end{cases}$$

i \ j	1	2	3	4	5
1					
2	x				
3	x	x			
4	x	x	x		
5	x	x	x	x	

Step 1: Fill the table for $i = j$

i \ j	1	2	3	4	5
1	0				
2	x	0			
3	x	x	0		
4	x	x	x	0	
5	x	x	x	x	0

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} & \text{otherwise} \end{cases}$$

i \ j	1	2	3	4	5
1	0				
2	x	0			
3	x	x	0		
4	x	x	x	0	
5	x	x	x	x	0

Step 2: Fill the table for:

$i=1, j=2$

$i=2, j=3$

$i=3, j=4$

$i=4, j=5$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

i \ j	1	2	3	4	5
1	0	120			
2	x	0	360		
3	x	x	0		
4	x	x	x	0	
5	x	x	x	x	0

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} \end{cases}$$

$$\begin{matrix} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & 10 \times 3 & 3 \times 12 & 12 \times 20 & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{matrix}$$

$$M[1,2] = \min_{1 \leq k < 2} \{M[1,1] + M[1+1,2] + p_0p_1p_2\}$$

$$M[1,2] = \min_{1 \leq k < 2} \{0 + 0 + 4 \times 10 \times 3\}$$

$$M[1,2] = 120$$

$$M[2,3] = \min_{2 \leq k < 3} \{M[2,2] + M[2+1,3] + p_1p_2p_3\}$$

$$M[2,3] = \min_{2 \leq k < 3} \{0 + 0 + 10 \times 3 \times 12\}$$

$$M[2,3] = 360$$

$$M[3,4] = \min_{3 \leq k < 4} \{M[3,3] + M[3+1,4] + p_2p_3p_4\}$$

$$M[3,4] = \min_{3 \leq k < 4} \{0 + 0 + 3 \times 12 \times 20\}$$

$$M[3,4] = 720$$

$$M[4,5] = \min_{4 \leq k < 5} \{M[4,4] + M[4+1,5] + p_3p_4p_5\}$$

$$M[4,5] = \min_{4 \leq k < 5} \{0 + 0 + 12 \times 20 \times 7\}$$

$$M[4,5] = 1680$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} & \text{otherwise} \end{cases}$$

$$\begin{array}{ccccccccc} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & & 10 \times 3 & & 3 \times 12 & & 12 \times 20 & & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{array}$$

i \ j	1	2	3	4	5
1	0	120			
2	x	0	360		
3	x	x	0	720	
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[1,3] = \min_{1 \leq k < 3}$$

$$k=1$$

$$= M[1,1] + M[1+1,3] + p_0p_1p_3$$

$$= 0 + 360 + 4 \times 10 \times 12$$

$$= 840$$

$$k=2$$

$$= M[1,2] + M[2+1,3] + p_0p_2p_3$$

$$= 120 + 0 + 4 \times 3 \times 12$$

$$= 264$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

i \ j	1	2	3	4	5
1	0	120	264		
2	x	0	360		
3	x	x	0	720	
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} & \text{otherwise} \end{cases}$$

$$\begin{matrix} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & 10 \times 3 & 3 \times 12 & 12 \times 20 & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{matrix}$$

$$M[2,4] = \min_{2 \leq k < 4}$$

$$k=2$$

$$= M[2,2] + M[2+1,4] + p_1p_2p_4$$

$$= 0 + 720 + 10 \times 3 \times 20$$

$$= 1320$$

$$k=3$$

$$= M[2,3] + M[3+1,4] + p_1p_3p_4$$

$$= 360 + 0 + 10 \times 12 \times 20$$

$$= 2760$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

i \ j	1	2	3	4	5
1	0	120	264		
2	x	0	360	1320	
3	x	x	0	720	
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} \end{cases}$$

$$\begin{array}{ccccccccc} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & & 10 \times 3 & & 3 \times 12 & & 12 \times 20 & & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{array}$$

$$M[3,5] = \min_{3 \leq k < 5}$$

$$k=3$$

$$= M[3,3] + M[3+1,5] + p_2p_3p_5$$

$$= 0 + 1680 + 3 \times 12 \times 7$$

$$= 1932$$

$$k=4$$

$$= M[3,4] + M[4+1,5] + p_2p_4p_5$$

$$= 720 + 0 + 3 \times 20 \times 7$$

$$= 1140$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} \end{cases}$$

$$\begin{array}{ccccccccc} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & & 10 \times 3 & & 3 \times 12 & & 12 \times 20 & & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{array}$$

i \ j	1	2	3	4	5
1	0	120	264		
2	x	0	360	1320	
3	x	x	0	720	1140
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[1,4] = \min_{1 \leq k < 4}$$

$$k=1$$

$$= M[1,1] + M[1+1,4] + p_0p_1p_4$$

$$= 0 + 1320 + 4 \times 10 \times 20$$

$$= 2120$$

$$k=2$$

$$= M[1,2] + M[2+1,4] + p_0p_2p_4$$

$$= 120 + 720 + 4 \times 3 \times 20$$

$$= 1080$$

$$k=3$$

$$= M[1,3] + M[3+1,4] + p_0p_3p_4$$

$$= 264 + 0 + 4 \times 12 \times 20$$

$$= 1224$$

Matrix Chain Multiplication

We want to start with $i = j$, then $i < j$ starting with a spread of 1, working our way up

$$M[i,j] = \begin{cases} 0 & \text{if } i=j \\ \min_{i \leq k < j} \{M[i,k] + M[k+1,j] + p_{i-1}p_kp_j\} \end{cases}$$

$$\begin{array}{ccccccccc} A_1 & \otimes & A_2 & \otimes & A_3 & \otimes & A_4 & \otimes & A_5 \\ 4 \times 10 & 10 \times 3 & 3 \times 12 & 12 \times 20 & 20 \times 7 \\ p_0 & p_1 & p_1 & p_2 & p_2 & p_3 & p_3 & p_4 & p_4 & p_5 \end{array}$$

i \ j	1	2	3	4	5
1	0	120	264	1080	
2	x	0	360	1320	
3	x	x	0	720	1140
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[2,5] = \min_{2 \leq k < 5}$$

$$k=2$$

$$= M[2,2] + M[2+1,5] + p_1p_2p_5$$

$$= 0 + 1140 + 10 \times 3 \times 7$$

$$= 1350$$

$$k=3$$

$$= M[2,3] + M[3+1,5] + p_1p_3p_5$$

$$= 360 + 1680 + 10 \times 12 \times 7$$

$$= 2880$$

$$k=4$$

$$= M[2,4] + M[4+1,5] + p_1p_4p_5$$

$$= 1320 + 0 + 10 \times 20 \times 7$$

$$= 2720$$

i \ j	1	2	3	4	5
1	0	120	264	1080	1344
2	x	0	360	1320	1350
3	x	x	0	720	1140
4	x	x	x	0	1680
5	x	x	x	x	0

$$M[1,5] = \min_{1 \leq k < 5}$$

$$k=1$$

$$= M[1,1] + M[1+1,5] + p_0 p_1 p_5$$

$$= 0 + 1350 + 4 \times 10 \times 7$$

$$= 1630$$

$$k=2$$

$$= M[1,2] + M[2+1,5] + p_0 p_2 p_5$$

$$= 120 + 1140 + 4 \times 3 \times 7$$

$$= 1344$$

$$k=3$$

$$= M[1,3] + M[3+1,5] + p_0 p_3 p_5$$

$$= 264 + 1680 + 4 \times 12 \times 7$$

$$= 2280$$

▶ Here dimensions are, $P_0 = 5, P_1 = 4, P_2 = 6, P_3 = 2, P_4 = 7$.

▶ There would be 4 matrices $A(5 \times 4), B(4 \times 6), C(6 \times 2)$ and $D(2 \times 7)$.

Example Solve the Following chain Matrix Multiplication

- ▶ Here dimensions are, $P_0 = 5, P_1 = 4, P_2 = 6, P_3 = 2, P_4 = 7$.
- ▶ There would be 4 matrices $A(5 \times 4), B(4 \times 6), C(6 \times 2)$ and $D(2 \times 7)$.

Longest Common Subsequence

- ▶ In biological applications, we often want to compare the DNA of two (or more) different organisms.
- ▶ A strand of DNA consists of a string of molecules called bases, where the possible bases are adenine, guanine, cytosine, and thymine (A, G, C, T).
- ▶ Comparison of two DNA strings
 - ❖ $S_1 = \text{ACCGGTCGAGTGCGCGGAAGCCGGCCGAA}$
 - ❖ $S_2 = \text{GTCGTTCGGAATGCCGTTGCTCTGTAAA}$
 - ❖ $S_3 = \text{GTCGTTCGGAAGCCGGCCGAA}$
- Similarity can be defined in different ways:
 - ❖ Two DNA strands are similar if one is a substring of the other.
 - ❖ Two strands are similar if the number of changes needed to turn one into the other is small.
 - ❖ There is a third strand S_3 in which the bases in S_3 appear in each of S_1 and S_2 ; these bases must appear in the same order, but not necessarily consecutively. The longer the strand S_3 we can find, the more similar S_1 and S_2 are. (we focus on this)

Longest Common Subsequence

- Similarity can be defined in different ways:
 - Two DNA strands are similar if one is a substring of the other.
 - Two strands are similar if the number of changes needed to turn one into the other is small.
 - There is a third strand S_3 in which the bases in S_3 appear in each of S_1 and S_2 ; these bases must appear in the same order, but not necessarily consecutively. The longer the strand S_3 we can find, the more similar S_1 and S_2 are. (we focus on this)

Longest Common Subsequence

Introduction

- ▶ A subsequence is a sequence that appears in the same relative order, **but not necessarily contiguous**.
- ▶ Given two sequences X and Y , we say that a sequence Z is a common subsequence of X and Y if Z is a subsequence of both X and Y .
- ▶ E.g., if $X = \langle \underline{A}, \underline{B}, C, B, \underline{D}, A, B \rangle$ and $Y = \langle \underline{B}, \underline{D}, C, A, \underline{B}, \underline{A} \rangle$ then $\langle B, C, A \rangle$ is a subsequence.
- ▶ Use dynamic programming technique to find **the longest common subsequence (LCS)**.

Longest Common Subsequence

- ▶ We need to generate table $c(1..m, 1..n)$
where m = length of string $S1$ and n = length of string $S2$.

To generate table $c[i][j]$ use following steps:

Step-1: Make $c[i][0] = 0$ and $c[0][j] = 0$

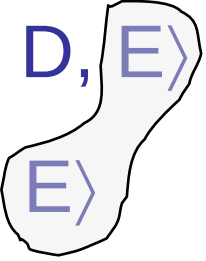
Step-2: if $x_i = y_j$ then $c[i, j] \leftarrow c[i - 1, j - 1] + 1$

Step-3: else $c[i, j] \leftarrow \max(c[i - 1, j], c[i, j - 1])$

A Recursive Solution

Case 1: $x_i = y_j$

e.g.: $X_i = \langle A, B, D, E \rangle$
 $Y_j = \langle Z, B, E \rangle$



$$c[i, j] = c[i - 1, j - 1] + 1$$

- Append $x_i = y_j$ to the LCS of X_{i-1} and Y_{j-1}
- Must find a LCS of X_{i-1} and $Y_{j-1} \Rightarrow$ optimal solution to a problem includes optimal solutions to subproblems

A Recursive Solution

Case 2: $x_i \neq y_j$

e.g.: $X_i = \langle A, B, D, G \rangle$

$Y_j = \langle Z, B, D \rangle$

$$c[i, j] = \max \{ c[i - 1, j], c[i, j - 1] \}$$

- Must solve two problems
 - find a LCS of X_{i-1} and Y_j : $X_{i-1} = \langle A, B, D \rangle$ and $Y_j = \langle Z, B, D \rangle$
 - find a LCS of X_i and Y_{j-1} : $X_i = \langle A, B, D, G \rangle$ and $Y_j = \langle Z, B \rangle$
- Optimal solution to a problem includes optimal solutions to subproblems

3. Computing the Length of the LCS

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

		0	1	2		n
		$y_j:$	y_1	y_2		y_n
0	x_i	0	0	0	0	0
1	x_1	0	→			
2	x_2	0	→			
		0			⋮	
		0				
m	x_m	0	→			

j

first
second
i

Additional Information

$$c[i, j] = \begin{cases} 0 & \text{if } i, j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

b & c:

	0	1	2	3	n
y_j :	A	C	D	F	
0 x_i	0	0	0	0	0
1 A	0				
2 B	0			$c[i-1, j]$	
3 C	0		$c[i, j-1]$		
	0				
m D	0				

j

i

A matrix $b[i, j]$:

- For a subproblem $[i, j]$ it tells us what choice was made to obtain the optimal value
- If $x_i = y_j$
 $b[i, j] = \nwarrow$
- Else, if
 $c[i-1, j] \geq c[i, j-1]$
 $b[i, j] = \uparrow$
 else
 $b[i, j] = \leftarrow$

LCS-LENGTH(X, Y, m, n)

```
1. for i ← 1 to m
2.   do c[i, 0] ← 0
3. for j ← 0 to n
4.   do c[0, j] ← 0
5. for i ← 1 to m
6.   do for j ← 1 to n
7.     do if  $x_i = y_j$ 
8.       then  $c[i, j] \leftarrow c[i - 1, j - 1] + 1$ 
9.          $b[i, j] \leftarrow \nwarrow$ 
10.    else if  $c[i - 1, j] \geq c[i, j - 1]$ 
11.      then  $c[i, j] \leftarrow c[i - 1, j]$ 
12.         $b[i, j] \leftarrow \uparrow$ 
13.    else  $c[i, j] \leftarrow c[i, j - 1]$ 
14.       $b[i, j] \leftarrow \leftarrow$ 
15. return c and b
```

The length of the LCS if one of the sequences is empty is zero

Case 1: $x_i = y_j$

Case 2: $x_i \neq y_j$

Running time: $\Theta(mn)$

Example

$X = \langle A, B, C, B, D, A \rangle$

$Y = \langle B, D, C, A, B, A \rangle$

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 \text{ or } j = 0 \\ c[i-1, j-1] + 1 & \text{if } x_i = y_j \\ \max(c[i, j-1], c[i-1, j]) & \text{if } x_i \neq y_j \end{cases}$$

If $x_i = y_j$

$b[i, j] = "$ ↖ "

Else if

$c[i-1, j] \geq c[i, j-1]$

$b[i, j] = "$ ↑ "

else

$b[i, j] = "$ ← "

		0	1	2	3	4	5	6
		Y_j	B	D	C	A	B	A
0	x_i	0	0	0	0	0	0	0
1	A	0	$\begin{matrix} \uparrow \\ 0 \end{matrix}$	$\begin{matrix} \uparrow \\ 0 \end{matrix}$	$\begin{matrix} \uparrow \\ 0 \end{matrix}$	$\begin{matrix} \nearrow \\ 1 \end{matrix}$	$\begin{matrix} \leftarrow \\ 1 \end{matrix}$	$\begin{matrix} \nearrow \\ 1 \end{matrix}$
2	B	0	$\begin{matrix} \nearrow \\ 1 \end{matrix}$	$\begin{matrix} \leftarrow \\ 1 \end{matrix}$	$\begin{matrix} \leftarrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \nearrow \\ 2 \end{matrix}$	$\begin{matrix} \leftarrow \\ 2 \end{matrix}$
3	C	0	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \nearrow \\ 2 \end{matrix}$	$\begin{matrix} \leftarrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$
4	B	0	$\begin{matrix} \nearrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \nearrow \\ 3 \end{matrix}$	$\begin{matrix} \leftarrow \\ 3 \end{matrix}$
5	D	0	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \nearrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 3 \end{matrix}$	$\begin{matrix} \uparrow \\ 3 \end{matrix}$
6	A	0	$\begin{matrix} \uparrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \nearrow \\ 3 \end{matrix}$	$\begin{matrix} \uparrow \\ 3 \end{matrix}$	$\begin{matrix} \nearrow \\ 4 \end{matrix}$
7	B	0	$\begin{matrix} \nearrow \\ 1 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 2 \end{matrix}$	$\begin{matrix} \uparrow \\ 3 \end{matrix}$	$\begin{matrix} \nearrow \\ 4 \end{matrix}$	$\begin{matrix} \uparrow \\ 4 \end{matrix}$

4. Constructing a LCS

- Start at $b[m, n]$ and follow the arrows
- When we encounter a “ $\nwarrow$ ” in $b[i, j] \Rightarrow x_i = y_j$ is an element of the LCS

		0	1	2	3	4	5	6
	y_j	B	D	C	A	B	A	
0	x_i	0	0	0	0	0	0	
1	A	0	↑ 0	↑ 0	↑ 0	↖ 1	← 1	↖ 1
2	B	0	↖ 1	← 1	← 1	↑ 1	↖ 2	← 2
3	C	0	↑ 1	↑ 1	↖ 2	← 2	↑ 2	↑ 2
4	B	0	↖ 1	↑ 1	↑ 2	↑ 2	↖ 3	← 3
5	D	0	↑ 1	↖ 2	↑ 2	↑ 2	↑ 3	↑ 3
6	A	0	↑ 1	↑ 2	↑ 2	↖ 3	↑ 3	↖ 4
7	B	0	↖ 1	↑ 2	↑ 2	↑ 3	↖ 4	↑ 4

Example 2

$X = \langle A, B, C, B, D, A, B \rangle$ and $Y = \langle B, D, C, A, B, A \rangle$ and $LCS = \langle B, D, A, B \rangle$

	y_j	A	B	C	B	D	A	B
x_i	0	0	0	0	0	0	0	0
B	0	↑ 0	↖ 1	← 1	↖ 1	← 1	← 1	↖ 1
D	0	↑ 0	↑ 1	↑ 1	↑ 1	↖ 2	← 2	← 2
C	0	↑ 0	↑ 1	↖ 2	← 2	↑ 2	↑ 2	↑ 2
A	0	↖ 1	↑ 1	↑ 2	↑ 2	↑ 2	↖ 3	← 3
B	0	↑ 1	↖ 2	↑ 2	↖ 3	← 3	↑ 3	↖ 4
A	0	↖ 1	↑ 2	↑ 2	↑ 3	↑ 3	↖ 4	↑ 4

Example 2

		C	B	D	A
	0	0	0	0	0
A	0	0	0	0	1
C	0	1	1	1	1
A	0	1	1	1	2
D	0	1	1	2	2
B	0	1	2	2	2

